

Online Appendix for:
LLMs as Economic Measurement Tools: Skill Demand, Workplace
Flexibility, and Firm Outcomes

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1 Classification Taxonomy

This appendix provides detailed definitions for all 25 classification categories used in our multi-label job posting classification framework. Categories are organized into three domains: digital skills, general skills, and workplace flexibility measures.

1.1 Digital Skills Categories

Category	Description
Artificial Intelligence (AI)	Skills related to AI technologies and machine intelligence.
Generative AI (GenAI)	Skills related to generating content using AI models.
CRM	Skills involving customer relationship management.
Computer Support and Networking	Skills in setting up, managing, and supporting computer systems and networks.
Data Analysis	Skills in analyzing, visualizing, and interpreting data.
Digital Design	Skills in creating digital content, including UI/UX and multimedia design.
Digital Marketing	Skills related to online marketing and analytics.
Machining and Manufacturing Technology	Skills related to manufacturing processes and technical engineering.
Productivity	Skills in tools that improve workflow and collaboration, like Excel.
Programming and Software Development	Skills in software programming, development, and related tools.

Table 1: Digital Skills Categories

1.2 General Skills Categories

Category	Description
Character Traits	Personal qualities or work habits that contribute to job performance.
Cognitive Skills	Intellectual and analytical skills for problem-solving.
Customer Service	Skills related to interacting with customers or clients.
Financial Skills	Skills in financial tasks like budgeting and accounting.
Management Skills	Skills required for team leadership, supervision, and resource management.
Social Skills	Interpersonal skills for effective communication and collaboration.
Writing and Language	Skills related to writing proficiency and language abilities.

Table 2: General Skills Categories

1.3 Workplace Flexibility Categories

Category	Description
Working from Home (wfhOptions)	Option to work outside a traditional office setting.
Company-Driven Flexibility Hours	Requires scheduling flexibility based on company needs.
Employee-Driven Flexibility Hours	Allows employees to set their own hours.
Flexible Tasks	Expectation for adaptability in tasks and responsibilities.
Temporary	Limited-duration or project-based positions.
Part Time	Position with reduced hours compared to full-time.
Shift Work	Work requiring non-standard or rotating shifts.
Travelling	Role requires regular travel or working across multiple locations.

Table 3: Workplace Flexibility Categories

2 Human Annotation Process

Following Ludwig et al. [2025]’s recommendation that LLM-based measurements require validation data, we construct a validation set by annotating 500 Danish and 500 English job postings, employing two native-speaker annotators (one per language). After removing job postings that were empty or failed quality checks, our final validation set contains 986 job postings.

2.1 Annotation Interface

Annotators used a web-based interface from Argilla for labeling job postings. The interface allowed:

1. Viewing the complete job posting text
2. Selecting category labels from dropdown menus
3. Highlighting relevant text spans
4. Adding evidence snippets for each positive classification
5. Quality control flags for ambiguous or problematic postings

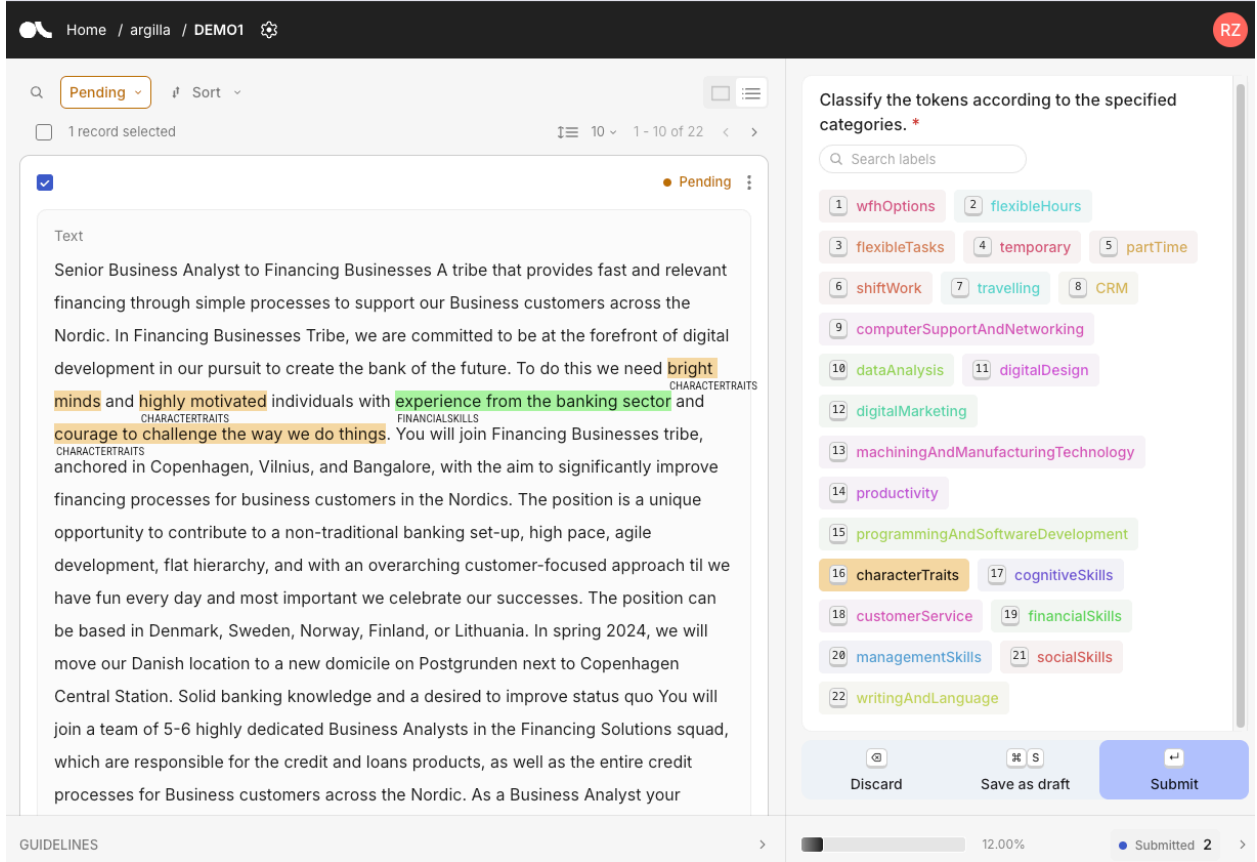


Figure 1: Human Annotation Interface Demo

2.2 Annotation Protocol

The process consisted of two stages: first, annotators identified key terms, phrases, or sentences related to skill and flexibility measures. Then, they labeled the identified text by selecting the appropriate category names and tagging the corresponding content, similar to using a marker. To maintain quality, annotators relied solely on domain knowledge and external source checks (e.g., Google search) for technical term clarifications, excluding AI-generated suggestions. This approach ensured consistent categorization, yielding a reliable validation set for evaluating different LLM optimization strategies.

3 Model Selection Criteria

3.1 Selected LLM Models

We evaluate models from the GPT-4.1 family:

- GPT-4.1: Flagship model, selected for state-of-the-art text understanding capabilities
- GPT-4.1-mini: Cost-effective variant with cost reduction and lower latency while maintaining comparable performance
- GPT-4.1-nano: Ultra-lightweight option

For open-source alternatives:

- Llama-3.3-70B: Large open-source model enabling on-premise deployment

All models support structured JSON output via response formatting, temperature control for deterministic outputs, and provide log probabilities for confidence assessment.

3.2 Open-Source vs Closed-Source Model Selection

The choice between open-source and closed-source models depends on data confidentiality requirements:

- Confidential Data: Use open-source models with local deployment. This includes administrative data, open-ended survey, and any data subject to privacy regulations.
- Non-Confidential Data: Use closed-source models with API access. This offers convenience and higher performance for publicly available data.

3.3 Model Deployment Considerations

LLMs are transformer-based neural networks that require substantial computational resources. Korinek [2024] provides detailed guidance on the practical aspects of LLM deployment, comparing API-based approaches with local deployment options and their respective

trade-offs for economic research applications. For public data without privacy constraints, researchers can access commercial LLM APIs directly. However, when data contains personal information or confidential records, models must be deployed on local infrastructure or secure servers.

This deployment constraint makes open-source models important for economic research. Our finding that the open-source model Llama-3.3-70B achieves performance parity with commercial APIs means researchers no longer face a trade-off between data security and classification quality.

Commercial API deployment involves non-trivial costs that scale with dataset size. For our full empirical application, classifying 4.3 million job postings using GPT-4.1-mini through batch API totaled \$3,447.

4 Keyword Baseline Details

4.1 Methodology

The keyword approach, a foundational text-as-data method in economics, analyzes text by aggregating words or phrases while disregarding word order. We implement a skill categorization framework using the comprehensive keyword list from Jensen [2020], Hassan et al. [2025], which builds on Deming and Kahn [2018]’s selective mapping strategy. Keywords are manually classified into skill categories or marked as noise, covering most occurrences. Synonyms from dictionary APIs extend this classification, with API definitions aiding uncategorized terms. Machine learning techniques further refine the process.

Skill extraction from job descriptions through keyword matching follows a three-stage process:

1. The text undergoes preprocessing through lowercasing and punctuation removal.
2. The preprocessed text is matched against a predefined keyword list to count skill mentions.

3. Skill measurement occurs by aggregating category frequencies and compiling matched keywords for each job posting.

4.2 Limitations

The keyword approach faces two fundamental limitations. First, it suffers from contextual blindness [Gentzkow et al., 2019, Hassan et al., 2025]. Since the keyword approach lacks semantic modeling, it can lead to mis-classification when terms have multiple meanings. For instance, “leverage” could indicate management skills when used as “leverage your position” or financial expertise when referring to “leverage ratio analysis.” Second, keyword methods can also generate false negatives. Job postings express requirements through varied vocabulary, which cannot fully be captured by predefined lists. These limitations motivate our LLM-based approach, which captures semantic meaning and identifies skills and flexibilities without being limited to a fixed list of terms.

4.3 Digital Skills Keyword Examples

Table 4: Digital Skills Categories with Bilingual Keyword Examples

Category	English Keywords	Danish Keywords
AI	machine learning, neural networks, tensorflow, pytorch, artificial intelligence, deep learning, ml, ai	maskinl�ring, neurale netv�erk, kunstig intelligens
GenAI	chatgpt, generative ai, prompt engineering, llm, large language model, gpt, openai	generativ ai, kunstig intelligens
CRM	salesforce, hubspot, crm software, customer relationship management	kundeservice system, crm
Computer Support	it support, cybersecurity, network administration, technical support, helpdesk, networking	it support, netv�erk, cybersikkerhed, teknisk support
Data Analysis	python, sql, tableau, data analysis, analytics, statistics, r programming, excel	dataanalyse, statistik, excel, python, sql
Digital Design	ui/ux, adobe suite, figma, photoshop, illustrator, graphic design, web design	grafisk design, webdesign, adobe, photoshop
Digital Marketing	seo, google analytics, social media marketing, digital marketing, sem, ppc	digital marketing, sociale medier, seo, online marketing
Manufacturing Tech	cnc, cad/cam, plcs, automation, manufacturing, machining	cnc, automatisering, produktion, maskiner
Productivity	microsoft office, project management tools, excel, powerpoint, word, outlook	microsoft office, excel, word, powerpoint, projektledele
Programming	programming, javascript, react, api development, software development, coding	programmering, softwareudvikling, kodning, javascript

4.4 General Skills Examples

Table 5: General Skills Categories with Bilingual Keyword Examples

Category	English Keywords	Danish Keywords
Character Skills	organized, reliable, adaptable, detail-oriented, self-motivated, proactive	organiseret, pålidelig, fleksibel, selvmotiveret, struktureret
Cognitive Skills	critical thinking, problem-solving, analytical, decision making, troubleshooting	problemløsning, analytisk, kritisk tænkning, fejlfinding
Customer Service	customer service, client relations, customer support, client management	kundeservice, kundekontakt, kundesupport, kundebehandling
Financial Skills	budgeting, financial analysis, accounting, finance, bookkeeping	økonomi, budgettering, regnskab, bogføring, finansiel analyse
Management	leadership, team management, supervision, project management	ledelse, teamledelse, projektledelse, lederskab, supervision
Social Skills	communication, teamwork, collaboration, interpersonal, team player	kommunikation, teamwork, samarbejde, sociale færdigheder
Writing/Language	writing, editing, language proficiency, content creation, technical writing	skrivning, sprogfærdigheder, kommunikation, tekstforfattning

4.5 Workplace Flexibility Examples

Table 6: Workplace Flexibility Categories with Bilingual Keyword Examples

Category	English Keywords	Danish Keywords
Work from Home	remote work, hybrid, telecommuting, work from home, wfh, remote	hjemmearbejde, remote arbejde, fleksibel arbejdsplads
Company Hours	on-call, rotating shifts, variable hours, flexible schedule, shift work	skiftarbejde, fleksible timer, varierende arbejdstid
Employee Hours	flexible hours, self-managed schedule, choose your hours, flexible timing	fleksible arbejdstider, selvbestemt arbejdstid, fri tilrettelæggelse
Flexible Tasks	varied responsibilities, cross-functional, diverse tasks, multiple duties	varierende opgaver, forskellige ansvarsområder, alsidige opgaver
Temporary Work	contract, seasonal, project-based, temporary, temp, short-term	kontrakt, midlertidig, projekt, sæsonarbejde, vikar
Part Time	part-time, reduced hours, part time, flexible hours	deltid, deltidsstilling, reducerede timer
Shift Work	shift work, night shift, evening shift, rotating shifts	skiftarbejde, natarbejde, aftenarbejde, weekendarbejde
Travelling	business travel, field work, travel required, on-site work	rejsearbejde, feltarbejde, kørsel, udstationering

5 LLM Implementation Details

5.1 Human-in-the-Loop Framework

Figure 2 illustrates our systematic approach for extracting economic variables from unstructured text, which combines human expertise with LLM capabilities through an iterative refinement process. The framework begins with a data classification decision (Step A) to determine whether to use open-source models with local deployment or closed-source models via API access, depending on data confidentiality requirements. After selecting the appro-

priate LLM (Step B), we construct a human annotation set to establish ground truth (Step C). In Step D, we send prompts to the AI and parse the cleaned output, then compare the AI results against human annotations. The critical innovation lies in the evaluation loop: if the LLM’s performance is acceptable, the process completes; otherwise, we employ three parallel optimization strategies: revising prompt structures, adjusting confidence thresholds based on false positive and negative analysis, and fine-tuning the model using validation data. This iterative approach ensures that our extraction method achieves satisfactory accuracy while maintaining scalability across different data sensitivity contexts.

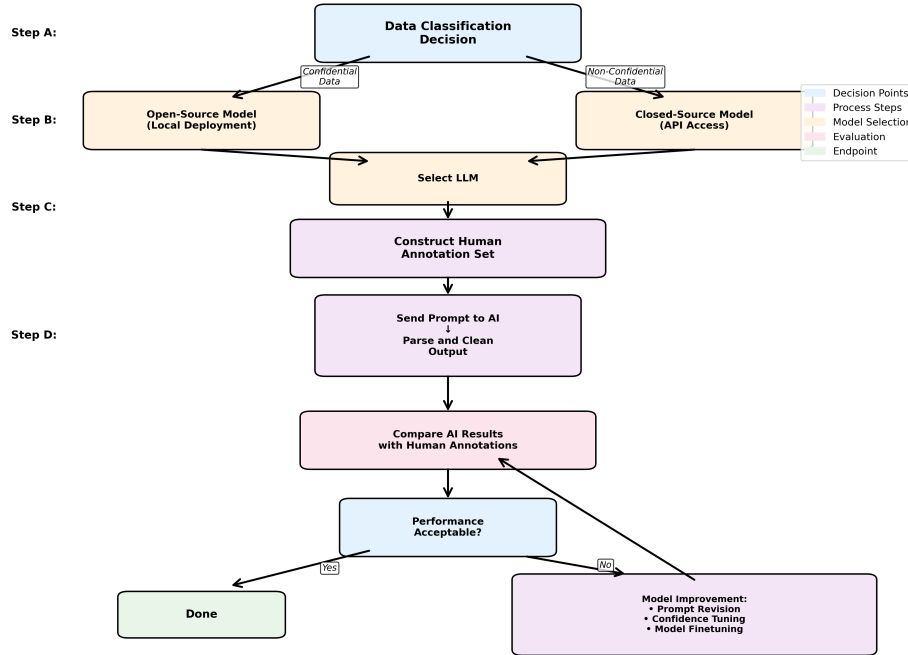


Figure 2: Human-in-the-Loop Framework for LLM-Based Multi-label Classification

5.2 Model Configuration and Structured Output

We implement both configurations: OpenAI models are accessed through their API, while open-source models are deployed locally on NVIDIA H100 GPUs. We configure the models with three settings:

1. We set temperature to 0, which means the model produces the least random outcomes when given the same input, minimizing randomness from predictions.

2. We enable the structured output function, which forces the model to return results in a format where each of the 25 categories receives a binary classification (1 for present, 0 for absent).
3. We enable log probabilities, which are numerical scores that indicate how confident the model is about each prediction. These confidence scores allow us to identify predictions where the model is uncertain, which we use later to improve classification accuracy.

5.3 Complete Base Prompt Text

Our baseline prompt employs a three-component structure for skill and flexibility measurement extraction. The prompt begins with task instructions specifying the dual objective of identifying required job skills and workplace flexibility arrangements. The core input section presents the complete job description text. The output specification mandates a structured JSON response format containing 25 binary classification categories.

Analyze this job posting to identify required skills and workplace flexibility options.

Job Description:
{job_description}

Return ONLY valid JSON in this exact format:

```
{
  "skills": {
    "AI": [1/0, "evidence"],
    "GenAI": [1/0, "evidence"],
    "CRM": [1/0, "evidence"],
    "computerSupportAndNetworking": [1/0, "evidence"],
    "dataAnalysis": [1/0, "evidence"],
    "digitalDesign": [1/0, "evidence"],
```

```

    "digitalMarketing": [1/0, "evidence"],
    "machiningAndManufacturingTechnology": [1/0, "evidence"],
    "productivity": [1/0, "evidence"],
    "programmingAndSoftwareDevelopment": [1/0, "evidence"],
    "characterSkills": [1/0, "evidence"],
    "cognitiveSkills": [1/0, "evidence"],
    "customerService": [1/0, "evidence"],
    "financialSkills": [1/0, "evidence"],
    "managementSkills": [1/0, "evidence"],
    "socialSkills": [1/0, "evidence"],
    "writingAndLanguage": [1/0, "evidence"]
  },
  "flexibility": {
    "wfhOptions": [1/0, "evidence"],
    "flexibilityHoursCompany": [1/0, "evidence"],
    "flexibilityHoursEmployee": [1/0, "evidence"],
    "flexibleTasks": [1/0, "evidence"],
    "temporary": [1/0, "evidence"],
    "partTime": [1/0, "evidence"],
    "shiftWork": [1/0, "evidence"],
    "travelling": [1/0, "evidence"]
  },
  "valid": [1/0, "reason if 0"]
}

```

Instructions:

- Return 1 if present, 0 if not
- Include key evidence phrases when found
- Set "valid" to 0 if not a job posting
- OUTPUT ONLY JSON, NO EXPLANATIONS OR REASONING

6 Prompt Engineering Strategies

We evaluate four prompt strategies designed to test different aspects of LLM classification behavior:

Table 7: Prompt Engineering Strategies

Strategy	Description
Binary	Direct 0/1 prediction for each category.
Binary with Evidence (Baseline)	0/1 prediction with supporting text snippets, enabling intensity measures.
Occurrence count	Extract category mentions instead of binary decisions, approximating human annotation.
Two-shot	Separate skill and flexibility classifications into distinct API calls to reduce task complexity.

Binary classification provides the simplest baseline by requesting 1/0 decisions for each category. Binary classification with evidence extends this by requiring supporting text snippets, enabling construction of within-category intensity measures from extracted phrases. Occurrence-based scoring asks models to extract category mentions rather than make binary decisions, attempting to replicate human annotation behavior. Two-shot decomposition separates skills and flexibility classifications into distinct API calls, examining whether reducing task complexity through decomposition improves performance or whether the integrated nature of job postings makes joint classification more effective.

The minimal performance differences across strategies indicate that prompt complexity has limited impact on classification outcomes. Binary classification achieves accuracy of 0.860 and macro F1 of 0.675, while the two-shot approach reaches nearly identical scores at 0.859 accuracy and 0.675 macro F1 despite doubling API calls. Binary classification with evidence shows slightly lower performance at 0.853 accuracy and 0.671 macro F1. Occurrence counting performs worst with 0.827 accuracy and 0.639 macro F1, indicating that occurrence-based approaches mis-align with the binary classification task. This paper adopts the binary classification with evidence strategy as the default approach to enable within-category intensity measurement through extracted text phrases.

7 Log Probability and Threshold Tuning

7.1 Confidence-Error Relationship

LLMs provide log probabilities representing token generation likelihood. Ivanova et al. [2024], Kauf et al. [2024] demonstrate that log probability scores measure model confidence and align predictions with human judgments. Values closer to zero indicate higher confidence. We use these scores to improve accuracy by filtering uncertain predictions through category-specific confidence thresholds that convert low-confidence positive predictions to negative ones, reducing false positives.

Our use of log probabilities to filter uncertain predictions also builds on Ma et al. [2025]’s analysis of LLM multi-label mechanisms, which shows that models exhibit spiky distributions with each step strongly favoring single labels. Their finding that relative label rankings poorly predict subsequent predictions motivates our category-specific threshold approach.

Figure 3 plots the relationship between category-level log probability scores and human-AI disagreement rates. Five categories exhibit high disagreement rates exceeding 20 percent: `cognitiveSkills`, `flexibleTasks`, `productivity`, `customerService`, and `managementSkills`. These categories show strong negative correlation between log probability values and disagreement rates, indicating that model uncertainty aligns with classification errors. Categories clustering near zero log probability with disagreement rates below 10 percent demonstrate reliable baseline performance. These include `wfhOptions`, `CRM`, and most digital skill categories.

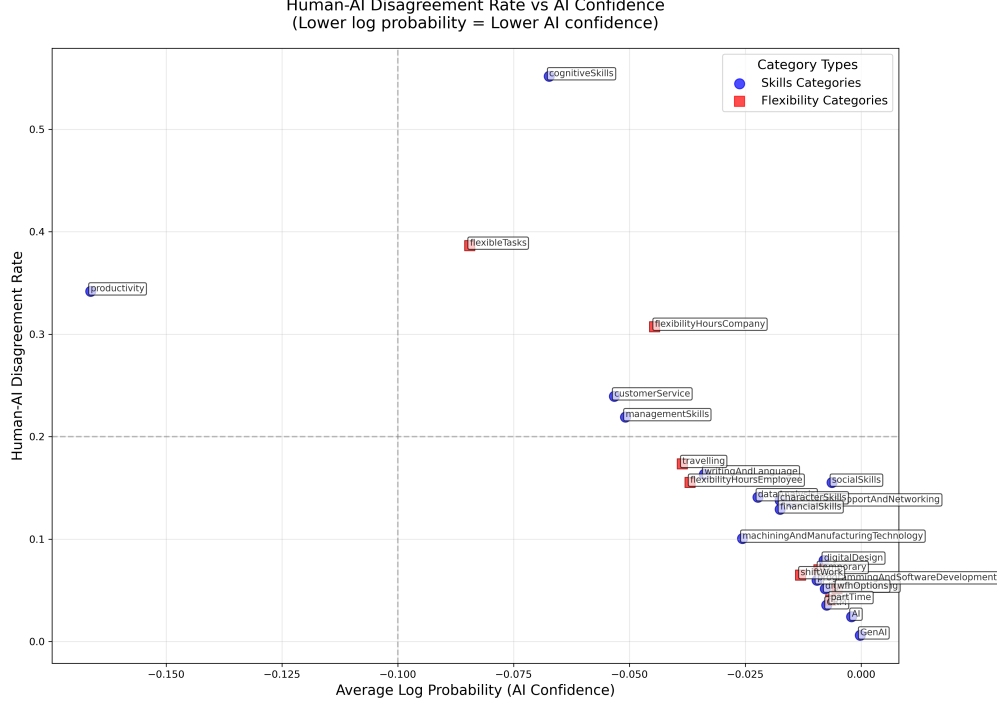


Figure 3: Disagreement vs. Log Probability by Category

7.2 Threshold Tuning Method

Based on the finding that model confidence correlates with classification errors, we develop a confidence-based threshold tuning method that exploits this relationship to improve measurement accuracy without additional training. Rather than accepting all positive predictions at face value, our method establishes category-specific confidence thresholds using log probabilities, converting uncertain positive predictions to negative classifications. This asymmetric treatment directly addresses the false positive problem that dominates classification errors.

The economic significance extends beyond performance metrics: threshold tuning enables researchers to calibrate measurement instruments for specific research objectives, prioritizing precision or recall. Categories with baseline error rates, such as flexible tasks and cognitive skills, show disagreement reductions exceeding 20 percentage points through threshold adjustment.

8 Fine-tuning Details

8.1 Overview

Fine-tuning LLMs adapts pre-trained language models to specific tasks by updating their internal neural network weights through additional training data [Howard and Ruder, 2018]. Unlike the base model trained on general text, fine-tuning adjusts the model’s parameters to optimize performance for the target classification task. This process preserves the model’s general language understanding while specializing its outputs for job posting classification.

8.2 Training Data Structure

The training data structure follows a format that mirrors human annotation practices. Each training example consists of an input-output pair where the input contains the complete job posting text followed by a template specification for each skill category, formatted as placeholder brackets indicating the expected output structure. The corresponding output contains the actual human annotations, with binary classifications (0 or 1) paired with evidence text extracted directly by human annotators when a skill is present. For instance, when annotators identify CRM skills in a posting, the output records both the positive classification and the specific text fragment supporting this judgment, such as “CRM ERP” or “Salesforce experience required.”

8.3 Implementation

We fine-tune the model gpt-4.1-2025-04-14 through OpenAI’s API. The training process employs three-fold cross-validation, with each fold using approximately 657 postings for training and 329 for testing.

Implementation complexity varies significantly between deployment methods:

- Commercial API fine-tuning through OpenAI requires only data formatting and API calls, completing training in approximately 2 hours per model.

- Open-source fine-tuning demands substantially more resources: configuring training infrastructure, managing GPU memory allocation, implementing gradient accumulation strategies, and monitoring convergence.

These technical requirements and computational costs make API-based fine-tuning more accessible for most research applications in the field of economics, though open-source alternatives remain necessary for sensitive and confidential data.

8.4 Model Improvement Techniques Summary

When base model performance is insufficient, we employ three parallel optimization strategies:

1. Prompt Revision: Iterative refinement of prompt structure and instructions to improve task clarity
2. Confidence Tuning: Threshold adjustment using log probability analysis to reduce false positives
3. Model Fine-tuning: Training specialized models on domain-specific annotated data

The choice among these techniques involves trade-offs between implementation complexity, computational cost, and expected performance gains. Prompt revision is cheapest but offers limited gains. Confidence tuning requires no additional training but works primarily for false-positive-dominated errors. Fine-tuning offers the largest performance improvements but requires substantial annotated data and computational resources.

9 Category-Level Performance

Tables 8 and 9 present complete performance metrics by category. Performances are highly heterogeneous. For categories such as `characterSkills` ($F1 = 0.916$) and `socialSkills` ($F1 = 0.906$), the model attains near-human performance. By contrast, some other categories such as `flexibleTasks` and `GenAI` fall below standard thresholds.

Trade-offs between precision and recall further reveal systematic classification biases. Categories with low accuracy, notably flexibleTasks, display high recall but low precision, consistent with systematic over-prediction. This motivates the threshold-tuning adjustment described in Section 7.

Table 8: Skill Category Performance (Sorted by F1 Score)

Category	Accuracy	Precision	Recall	F1
characterSkills	0.930	0.897	0.936	0.916
socialSkills	0.921	0.884	0.929	0.906
writingAndLanguage	0.891	0.845	0.876	0.860
programmingAndSoftwareDevelopment	0.952	0.873	0.831	0.851
dataAnalysis	0.943	0.857	0.818	0.837
productivity	0.762	0.696	0.891	0.782
cognitiveSkills	0.729	0.683	0.856	0.760
managementSkills	0.788	0.701	0.812	0.753
customerService	0.814	0.723	0.789	0.754
financialSkills	0.924	0.756	0.712	0.733
digitalMarketing	0.961	0.789	0.652	0.714
digitalDesign	0.967	0.812	0.623	0.705
CRM	0.978	0.845	0.567	0.679
computerSupportAndNetworking	0.956	0.798	0.534	0.640
machiningAndManufacturingTechnology	0.962	0.756	0.489	0.594
AI	0.982	0.823	0.412	0.549
GenAI	0.991	0.756	0.287	0.416

Table 9: Flexibility Category Performance (Sorted by F1 Score)

Category	Accuracy	Precision	Recall	F1
partTime	0.967	0.912	0.867	0.889
temporary	0.954	0.878	0.845	0.861
shiftWork	0.943	0.856	0.823	0.839
travelling	0.912	0.834	0.801	0.817
wfhOptions	0.934	0.812	0.789	0.800
flexibilityHoursEmployee	0.878	0.756	0.734	0.745
flexibilityHoursCompany	0.856	0.723	0.712	0.717
flexibleTasks	0.689	0.534	0.878	0.664

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10 Confidence as an Error Detector

The confidence scores used for recalibration also serve as an audit signal. On the validation set, confidence ranks classification disagreement with an AUC of 0.80; auditing the least-confident 20% of predictions uncovers 50% of all errors, and restricting to the most-confident half of predictions raises accuracy from 0.85 to 0.97. Across the 25 substantive categories, mean confidence rank-orders eventual human-AI disagreement with a Spearman correlation of -0.92 , allowing researchers to allocate annotation budgets to the categories that need validation before any labels are collected.